

When a nonnegative scalar symbol acquires a negative part

A nonnegative function on phase space need not quantize to a nonnegative operator. Nevertheless its negative quadratic form can be bounded by a constant when the symbol has the appropriate size relative to its derivative scale. For scalar symbols the permitted size is the inverse square of the local Planck parameter. The extra power comes from a decomposition of a nonnegative function into a square and a function missing one phase direction. This is a scalar argument; it is not a theorem about positive matrices or arbitrary Hilbert space operators.

We will build the estimate around two operations that preserve lower bounds: taking an operator square and integrating a lower-dimensional estimate with a parameter. A separate localization lemma accounts for the errors when these local constructions are assembled. The small-scale construction and the subsequent passage to an arbitrary metric are different steps of the proof.

Use the Fourier convention $D = -i\partial$, Lebesgue measure on $\mathbb{R}^n$, and

$$a^w u(x) = (2\pi)^{-n} \iint e^{i(x-y)\cdot\xi} a((x+y)/2, \xi) u(y) dy d\xi. \quad (\text{F1})$$

The integral is interpreted by the Schwartz-distribution construction of Section 4 of Two measuring scales, one Weyl product and Section 4 of From Weyl symbols to operators and changes of coordinates. All operator inequalities below are quadratic-form inequalities on $\mathcal{S}(\mathbb{R}^n)$. They do not assert that an unbounded operator has already been given a self-adjoint realization.

The proof uses four kinds of prerequisites.

- **Metric constructions.** Sections 1–4 of Localizing symbols with moving metrics give the symbol seminorms, slowly varying ellipsoid covers, uniformly differentiable cutoffs, and reconstruction. Sections 1–3 of Two measuring scales, one Weyl product give symplectic duality, temperate weights, and the Planck parameter. Section 5 of From Weyl symbols to operators and changes of coordinates gives the conformal enlargement theorem, including nonsmooth metrics.
- **Weyl calculus.** Section 7 of Two measuring scales, one Weyl product gives the Weyl product and every finite remainder for compatible temperate metrics. For one metric the remainder after terms of order less than N has weight $m_1 m_2 h^N$. Section 4 of From Weyl symbols to operators and changes of coordinates and Section 6 of From Weyl symbols to operators and changes of coordinates give Schwartz action and unitary affine symplectic covariance. Section 7 of From Weyl symbols to operators and changes of coordinates gives change of quantization with its finite remainders.
- **Operator bounds.** The general-metric continuity theorem in Section 7 of When a moving symbol scale controls an operator supplies the following precise interface: for a slowly varying symplectically temperate metric $g \leq g^\sigma$, a symbol in $S(1, g; \mathcal{L}(H_1, H_2))$ defines a bounded operator on L^2 , with a norm controlled by finitely many symbol seminorms and the metric's structural constants. Constants are independent of the Hilbert spaces and of pointwise eccentricity. Section 1 of When a moving symbol scale controls an operator supplies the operator-norm version of the preceding Weyl product and every finite remainder estimate, preserving multiplication order and matching intermediate coefficient spaces. Constant metrics have uniform structural constants. The boundedness result used here supplies no positivity theorem.
- **Elementary analytic tools.** The metric and calculus lesson, Sections 12–14, proves compactness and extrema, complete Taylor/product/chain formulas and smooth cutoffs. The Fourier lesson, Sections 1–3 and 7–8, proves scalar Fourier inversion, Plancherel, distribution duality and both real self-adjoint and skew-adjoint spectral decompositions with the original inner product. The geometric lesson, Sections 16.4–16.5, proves the full inverse and implicit maps. The Banach and

measure lesson, Sections 16–17, proves product integration, dominated convergence and integration of Hilbert-valued functions without an ambient separability assumption. These are the exact independent providers; the splitting and induction arguments for positivity follow below.

1. The scale in the two lower bounds

On $E = \mathbb{R}_x^n \times \mathbb{R}_\xi^n$ use $\sigma((x, \xi), (y, \eta)) = \xi \cdot y - x \cdot \eta$. For a positive quadratic form g_X , put

$$g_X^\sigma(T) = \sup_{S \neq 0} \frac{|\sigma(T, S)|^2}{g_X(S)}, \quad h(X)^2 = \sup_{T \neq 0} \frac{g_X(T)}{g_X^\sigma(T)}. \quad (\text{F2})$$

Here a permissible metric means a slowly varying, symplectically temperate metric satisfying $h \leq 1$. Symplectic temperateness has the distance base specified in Section 2 of Two measuring scales, one Weyl product; for example it can be written

$$g_Y \leq C g_X (1 + g_Y^\sigma(X - Y))^N. \quad (\text{F3})$$

No ordinary smoothness of $X \mapsto g_X$ is assumed. The seminorms are the multilinear derivative seminorms of Section 1 of Localizing symbols with moving metrics. In particular $a \in S(h^{-j}, g)$ means

$$|a^{(k)}(X)[T_1, \dots, T_k]| \leq C_k h(X)^{-j} \prod_{l=1}^k g_X(T_l)^{1/2}. \quad (\text{F4})$$

The Planck parameter and its powers are temperate weights. To verify the point needed here, compare g_X with g_Y in (F3), dualize that comparison, and take the supremum defining (F2). The resulting ratio of Planck parameters is bounded by a power of the same distance. Local comparisons follow by the same argument from slow variation.

Both estimates here concern scalar symbols. Our two conclusions are

$$\begin{aligned} 0 \leq a \in S(h^{-1}, g) &\implies \langle a^w u, u \rangle \geq -C \|u\|^2, \\ 0 \leq a \in S(h^{-2}, g), \quad a \text{ scalar} &\implies \langle a^w u, u \rangle \geq -C \|u\|^2. \end{aligned} \quad (\text{F5})$$

The first will follow from one adapted square root. The second, the scalar Fefferman–Phong estimate, will follow from dimension induction. Constants depend on finitely many seminorms of a , the dimension and the structural constants of g . They are uniform on sets on which these data are bounded. The proof will explain why finitely many derivatives suffice; no optimal derivative count is claimed.

2. A square root measured at its own scale

We first prove the first line of (F5). A basic positivity estimate will be useful. If $F \geq 0$ on a fixed ball, F and its second derivatives are bounded there, then on a smaller concentric ball

$$|F'(z)| \leq C F(z)^{1/2}. \quad (\text{F6})$$

For a unit direction v , Taylor’s formula gives $0 \leq F(z + tv) \leq F(z) + tF'(z)v + Ct^2$ in both signs of t . Choosing the sign opposite to $F'(z)v$ gives $|F'(z)v| \leq F(z)/s + Cs$ for every allowed $s > 0$. If $F(z)$ is small choose s proportional to $F(z)^{1/2}$; otherwise use a fixed s and the boundedness of F . The case $F(z) = 0$ follows by letting $s \downarrow 0$. Taking the supremum over directions proves (F6).

Let $a \geq 0$ belong to $S(h^{-1}, g)$, and set

$$m = a + 1, \quad G_X = \frac{g_X}{h(X)m(X)}. \quad (\text{F7})$$

At a fixed point X , choose g_X -orthonormal coordinates z centered at X . Slow variation of g, h makes $F(z) = h(X)a(X+z)$ and all its derivatives uniformly bounded on a fixed coordinate ball. Applying (F6) at its center gives

$$|a'(X)T| \leq C\sqrt{a(X)/h(X)}g_X(T)^{1/2}. \quad (\text{F8})$$

For $k \geq 2$, the original symbol estimate implies

$$|a^{(k)}(X)[T_1, \dots, T_k]| \leq C_k m(X)^{1-k/2} h(X)^{-k/2} \prod_l g_X(T_l)^{1/2}. \quad (\text{F9})$$

Indeed hm is bounded above, and division of the right side by $h^{-1} \prod g_X(T_l)^{1/2}$ gives $(hm)^{1-k/2}$, bounded below by a positive constant. For $k = 1$, (F9) follows from (F8). Thus $m \in S(m, G)$.

We check the metric assumptions rather than assuming a smooth square root has already solved the problem. In the coordinates just used,

$$|F(z) - F(0)| \leq C\sqrt{F(0)}|z| + C|z|^2. \quad (\text{F10})$$

If $|z| \leq c\sqrt{F(0) + h(X)}$, with c fixed sufficiently small, this is at most $(F(0) + h(X))/2$. Such a displacement lies inside the original coordinate ball since hm is uniformly bounded. Consequently $a(X+z) + 1$ and $a(X) + 1$ are comparable, as are g, h . This is exactly slow variation of G . Quadratic scaling gives

$$h_G = \frac{1}{m} \leq 1. \quad (\text{F11})$$

The original conformal multiplier in (F7) satisfies the explicit bound

$$\mu(X) = \frac{1}{h(X)m(X)} \geq \mu_0, \quad \mu_0 = \frac{1}{1 + p_0(a; h^{-1}, g)} > 0, \quad (\text{F11a})$$

because $h(X)a(X) \leq p_0(a; h^{-1}, g)$ and $h(X) \leq 1$. Keep the original g and this exact G . Equations (A56)–(A58) of From Weyl symbols to operators and changes of coordinates apply with precisely (F11a), the slow variation just proved, and (F11). They prove temperateness of G , including the distance base and dependence on μ_0 . No metric is replaced. This argument does not require the multiplier or h to be smooth. Equation (F11), with the Planck-parameter weight comparison proved after (F4), also proves temperateness of m for G .

For completeness, if $M > 0$ is a smooth symbol in $S(M, G)$ and a positive smooth scalar function F satisfies $|t^j F^{(j)}(t)| \leq C_j F(t)$, the chain rule indexed by set partitions gives

$$D^k(F(M)) = \sum_{\pi} F^{(|\pi|)}(M) \prod_{B \in \pi} D^{|B|} M. \quad (\text{F12})$$

The powers of M cancel in each term. The bound on the first logarithmic derivative also gives $|\log F(t) - \log F(s)| \leq C|\log(t/s)|$, so local comparison and temperateness of M imply the same properties for $F(M)$. This proves $F(M) \in S(F(M), G)$. In particular $b = \sqrt{m} \in S(m^{1/2}, G)$. The first Weyl correction of $b \# b$ vanishes because the Poisson bracket of a scalar function with itself is zero. The order-two remainder gives

$$b \# b = m + r, \quad r \in S(mh_G^2, G) = S(m^{-1}, G) \subset S(1, G). \quad (\text{F13})$$

The operator r^w is bounded by Section 7 of When a moving symbol scale controls an operator. Since b is real, $\langle (b^w)^2 u, u \rangle = \|b^w u\|^2$ on Schwartz functions. Thus $\langle a^w u, u \rangle = \|b^w u\|^2 - \|u\|^2 - \langle r^w u, u \rangle$, proving the first line of (F5).

3. What four derivatives and nonnegativity control

Here $B_r \subset \mathbb{R}^d$ is the Euclidean ball of radius r . For this finite-dimensional lemma only, write $j_k(f, x) = \sup_{|v|=1} |D^k f(x)[v, \dots, v]|$. These diagonal seminorms are equivalent to multilinear norms by polarization; all constants below may depend on d, k . For $k \leq 3$, real symmetric derivatives have exactly the same diagonal and multilinear norms. Thus the numerical bounds below also hold in the multilinear convention of Section 1 of Localizing symbols with moving metrics. The cases $k \leq 1$ are immediate, and $k = 2$ is the spectral norm identity for a real symmetric matrix.

Here is a finite-dimensional proof for $k = 3$, to retain the numerical constants. Let T be a real symmetric trilinear form and M its multilinear norm. The zero form is immediate. Among unit triples attaining $|T(x, y, z)| = M$, choose one maximizing $\|x + y + z\|$, using compactness. Fix z and let B be the symmetric operator with $\langle Bx, y \rangle = T(x, y, z)$. At an extremal triple, differentiation on each unit sphere gives $Bx = \varepsilon My$ and $By = \varepsilon Mx$, where ε is the sign of $T(x, y, z)$. If $0 < r = \|x + y\| < 2$, set $v = (x + y)/r$. Then $Bv = \varepsilon Mv$, so (v, v, z) is another maximizing triple. Its squared sum norm exceeds the previous one by

$$(2 - r)(2 + r + 2\langle v, z \rangle) \geq (2 - r)r > 0,$$

a contradiction. If $x = -y$, replacing the pair by (v, v) , where v is either x or $-x$ and $\langle v, z \rangle \geq 0$, preserves the absolute trilinear value and increases the squared sum norm from one to at least five. Hence $x = y$. Applying the same argument to each pair shows $x = y = z$, which proves equality with the diagonal norm. At order four we only need that a multilinear bound implies the same diagonal bound, and that polarization controls mixed derivatives by a fixed dimensional constant.

Normalized jet lemma. Suppose $f \geq 0$ is smooth on B_2 , $j_4(f, x) \leq 1$ there, and

$$\max\{f(0), j_2(f, 0)\} = 1. \tag{F14}$$

There is a radius $r > 0$, independent of f , such that on B_r

$$\frac{1}{2} < \max\{f(x), j_2(f, x)\} < 2, \quad j_k(f, x) < 8 \quad (0 \leq k < 4). \tag{F15}$$

For $k = 0$ use $j_0 = f$.

Proof. Fix a unit vector v , put $L = Df(0)v$, $T = D^3f(0)[v, v, v]/6$, and evaluate Taylor's formula at $\pm v$ and at $\pm 2v$, using limits from inside B_2 for the latter. Nonnegativity and (F14) imply

$$|L + T| \leq \frac{37}{24}, \quad |2L + 8T| \leq \frac{11}{3}. \tag{F16}$$

Subtracting twice the first expression from the second gives $|6T| \leq 27/4 < 7$. Subtracting one quarter of the second from twice the first gives $|3L/2| \leq 4$, hence $|L| \leq 8/3 < 3$. Therefore the diagonal third and first derivatives at zero are uniformly bounded, with strict room below eight. Polarization bounds their mixed versions. Taylor's formula now bounds the change of the Hessian by $C|x|$, the change of the first derivative by $C|x|$, and the change of the third derivative by $C|x|$, using the fourth derivative bound. The same holds for f . Choose a single small radius so that these changes preserve the strict inequalities in (F15). One of the two quantities in (F14) equals one, which gives its lower bound as well. $\square$

The argument also gives a version without normalization: if $f(0), j_2(f, 0) \leq 1$ and $j_4 \leq 1$, all derivatives through order three are uniformly bounded on a fixed smaller ball. Apply the preceding proof to $f + 1 - f(0)$, which remains nonnegative and has value one at zero. This variant gives upper bounds; it does not produce a nonnegative splitting of the original f by subtracting the added constant.

4. Removing one phase direction by a scalar splitting

Splitting lemma. Under (F14), there is a fixed smaller radius on which

$$f(s, y) = v(y) + q(s, y)^2, \quad v \geq 0, \quad (\text{F17})$$

after an orthogonal coordinate change. Here s is one real direction, v is independent of it, and q, v are smooth and real. Their derivatives through order k are controlled by finitely many bounds for f through order $k + 2$ on a fixed larger working ball. Both balls and all constants are independent of f when these bounds are fixed. In particular no nondegeneracy assumption on the entire Hessian is imposed.

Proof. The jet lemma first gives uniform bounds through order three on a fixed ball. We choose radii and a positive threshold η using only those bounds.

If $f(0) \geq \eta$, shrink the output ball until $f \geq \eta/2$, using the uniform first derivative bound. Set $v = 0$, choose any direction, and take $q = \sqrt{f}$. Repeated chain differentiation controls q to order k by derivatives of f to order k and the fixed positive lower bound.

Suppose instead $f(0) < \eta$. Then the Hessian has spectral norm one. It has an eigenvalue of absolute value one. Such an eigenvalue cannot be -1 when $\eta \leq 1/16$: for a corresponding unit vector, the average of Taylor's formulas at $v/2$ and $-v/2$ would give

$$0 \leq \frac{f(v/2) + f(-v/2)}{2} \leq f(0) - \frac{1}{8} + \frac{1}{384} < 0. \quad (\text{F18})$$

The odd terms cancel in this test. Hence there is an eigenvector with eigenvalue $+1$. Choose it as the s direction. At the origin $f_{ss} = 1$ and $f_{sy_j} = 0$. The third derivative bound gives $f_{ss} \geq 1/2$ on a fixed ball B_R .

By (F6), $|f_s(0)| \leq C\sqrt{f(0)}$. Taylor expansion in y , and the vanishing mixed Hessian at the origin, give

$$|f_s(0, y)| \leq C\sqrt{\eta} + C|y|^2. \quad (\text{F19})$$

Choose a transverse radius smaller than $R/4$, then choose it smaller if needed, and finally choose η small enough so that this bound is less than $R/8$. On the line segment $|s| \leq R/2$ the derivative $f_s(s, y)$ is strictly increasing, and has opposite signs at its endpoints. There is a unique zero $s = X(y)$ in that segment. The implicit function theorem gives a smooth X . The whole graph and every segment between it and the output ball remain inside B_R , by the chosen margins.

Set $v(y) = f(X(y), y) \geq 0$. Taylor expansion about this critical point is the exact identity

$$f(s, y) = v(y) + (s - X(y))^2 A(s, y), \quad A(s, y) = \int_0^1 (1 - t) f_{ss}(X(y) + t(s - X(y)), y) dt. \quad (\text{F20})$$

The coefficient $A \geq 1/4$; take $q = (s - X(y))\sqrt{A}$. To track regularity, differentiate $f_s(X(y), y) = 0$. The term containing a derivative of X of order k has coefficient $f_{ss} \geq 1/2$; every other term involves a derivative of X of lower order and a derivative of f of order at most $k + 1$. Induction therefore bounds X through order k . Differentiating the integral in (F20) uses derivatives of f through order $k + 2$. Its fixed positive lower bound permits the square-root chain rule. This proves the stated bounds for q , and the chain rule for $f(X(y), y)$ proves those for v . $\square$

The function v can have several zeros in the transverse neighborhood, and the critical graph need not be linear. What matters is that v is independent of the *constant* direction ∂_s . The decomposition is not a change of phase coordinates in the quantization; later only a linear symplectic rotation of that direction will be used.

5. Squared localization and its two-derivative error

Let g be permissible and $m = h^{-2}$. Choose real smooth functions ϕ_ν , supported in permissible metric balls, with

$$\sum_\nu \phi_\nu^2 = 1, \quad (\phi_\nu)_\nu \text{ bounded in } S(1, g; \ell^2). \quad (\text{F21})$$

They are obtained from the ellipsoid cover of Section 2 of Localizing symbols with moving metrics: take cutoffs θ_ν equal to one on its smaller covering balls and set $\phi_\nu = \theta_\nu / (\sum_\mu \theta_\mu^2)^{1/2}$. The denominator is bounded below by one. Only a fixed number of terms occur at each point, and the product and chain rules give every seminorm in (F21). This also proves boundedness of all derivatives of the column in ℓ^2 , without a factor depending on the number of balls.

Suppose real symbols a_ν are supported in fixed slightly larger balls, have uniform $S(m, g)$ seminorms there, and satisfy $\phi_\nu^2 a_\nu = \phi_\nu^2 a$. Then

$$\sum_\nu \|\phi_\nu^w u\|^2 \leq C \|u\|^2, \quad \sum_\nu \phi_\nu^w a_\nu^w \phi_\nu^w = a^w + R^w, \quad R \in S(1, g). \quad (\text{F22})$$

The sum in the second identity is interpreted weakly on Schwartz functions. The norm of the remainder uses only finitely many input seminorms. The finite-derivative theorem below applies this identity to compactly supported approximants before taking its final limit.

The first estimate applies Section 7 of When a moving symbol scale controls an operator to the column in (F21). For a finite index set, apply the operator-norm Weyl product to that column, the diagonal matrix with entries a_ν , and its real row. Finite overlap gives uniform symbol bounds for the diagonal as well. The zeroth term is $\sum \phi_\nu^2 a_\nu$. For each scalar entry the first terms cancel:

$$\{\phi_\nu, a_\nu\} \phi_\nu + \{\phi_\nu a_\nu, \phi_\nu\} = 0. \quad (\text{F23})$$

Expanding the two products to order two, and using the product rule on the first correction, shows that all remaining terms have weight $mh^2 = 1$. The constants involve only finitely many input seminorms and the fixed overlap. This calculation uses the scalar nature of ϕ_ν ; it does not discard a noncommutative first correction between two arbitrary operator-valued symbols.

For an exhaustion by finite sets, embed all finite columns and diagonal matrices into the same space ℓ^2 by inserting zero entries. Local finiteness of the larger supports makes these symbols converge locally smoothly in operator norm to their full column and diagonal, with uniform symbol seminorms. The bounded-set local-smooth continuity in Section 1 of When a moving symbol scale controls an operator therefore makes their Weyl products and each finite remainder converge in the corresponding symbol classes in this sense. The principal sums converge locally smoothly to a , and the remainders converge to a symbol $R \in S(1, g)$. Polynomial control from temperateness makes these bounded local convergences distributional. The Weyl kernel pairing with Schwartz functions passes to the limit and proves (F22), including its weak sum interpretation.

In particular, if every localized operator obeys $\langle a_\nu^w v, v \rangle \geq -C_0 \|v\|^2$ with the same C_0 , the finite identities followed by this limit give

$$\langle a^w u, u \rangle \geq -C \|u\|^2. \quad (\text{F24})$$

This implication uses no positivity of the quantized cutoffs themselves. It uses their squared L^2 estimate, the local lower bounds, and a bounded remainder.

6. A metric selected by the value and Hessian

Consider a nonnegative smooth function a on $\mathbb{R}^{2n}$ satisfying, for $0 < \lambda \leq 1$,

$$|a^{(k)}(X)| \leq \lambda^{(k-4)/2} \quad (0 \leq k \leq N), \quad (\text{F25})$$

where the norms are Euclidean multilinear norms and the integer N is at least four. Set

$$L(X) = \max\{1, \sqrt{a(X)}, |a''(X)|\}, \quad H(X) = L(X)^{-1}, \quad G_X = H(X)e. \quad (\text{F26})$$

Then $\lambda \leq H \leq 1$ by the bounds at orders zero and two. We prove that G is permissible with uniform structural constants, and that the localized seminorms of a in $S(H^{-2}, G)$ through order N are uniformly bounded. The nonsmooth maximum in (F26) causes no difficulty for the definition of a metric.

At a center X , use the rescaled function

$$f_X(z) = H(X)^2 a(X + H(X)^{-1/2} z). \quad (\text{F27})$$

It has value and Hessian norm at zero at most one. Its fourth derivative is bounded by one. For $k \geq 4$, (F25) gives

$$|f_X^{(k)}(z)| \leq (\lambda/H(X))^{(k-4)/2} \leq 1. \quad (\text{F28})$$

The unnormalized variant of the jet lemma gives uniform bounds for its lower derivatives on a fixed small ball. Consequently, when $|Y - X|/\sqrt{H(X)}$ is small, $L(Y) \leq 2L(X)$, after decreasing the radius to make the bounds for the value and Hessian smaller than two.

If $H(X) < 1$, at least one of $f_X(0)$ and $|f_X''(0)|$ equals one. The lower bound of (F15) then gives $L(Y) \geq L(X)/2$ on a fixed smaller ball. If $H(X) = 1$, this lower bound follows directly from $L(Y) \geq 1$. We have proved both sides of slow variation, with constants independent of a, λ . The same argument and (F28) give the asserted derivative bounds after returning to the original coordinates.

Quadratic duality gives $G_X^\sigma = H(X)^{-1}e$ and $h_G = H \leq 1$. Temperateness here has a particularly short verification. If $G_X(X - Y)$ is within the slow-variation radius, use local comparison. Otherwise $H(X)|X - Y|^2 \geq c > 0$, and, since $H(Y) \leq 1$,

$$\frac{H(Y)}{H(X)} \leq \frac{1}{c} \frac{|X - Y|^2}{H(Y)}. \quad (\text{F29})$$

Together the two cases prove (F3) for G with fixed constants. In particular H^{-2} is a temperate weight with fixed constants.

Choose the squared partition of Section 5 for G , with supports in balls small enough for (F27) and the splitting lemma. Let χ be a fixed cutoff equal to one on the smaller partition support, with support inside the splitting ball, and set

$$a_\nu(Y) = \chi(\sqrt{H_\nu}(Y - X_\nu))^2 a(Y), \quad H_\nu = H(X_\nu). \quad (\text{F30})$$

The symbols a_ν have uniform seminorms in both their frozen classes $S(H_\nu^{-2}, H_\nu e)$ and the moving class $S(H^{-2}, G)$, to the controlled finite order. These statements follow by the product rule, (F27)–(F28), and local comparison; the functions are zero off their larger balls. No derivative of H is taken.

7. The uniform estimate for every constant metric

Theorem. For each spatial dimension n , there are an integer N_n and a constant C_n with the following property. Let g be any constant positive quadratic form with $h_g \leq \lambda \leq 1$. If $a \geq 0$ is scalar and

$$|a|_{k,g} \leq \lambda^{-2} \quad (0 \leq k \leq N_n), \quad (\text{F31})$$

then $\langle a^w u, u \rangle \geq -C_n \|u\|^2$. The constants are independent of g, a, λ .

Symplectic normalization. A positive quadratic form has symplectic coordinates in which it is $\sum_j \lambda_j (dx_j^2 + d\xi_j^2)$, with $\lambda_j > 0$ and $\max_j \lambda_j = h_g$. Here is a direct construction. In a g -orthonormal basis, the matrix of the symplectic form is real, invertible and skew-adjoint. Orthogonal spectral decomposition produces orthogonal pairs v_j, w_j with $\sigma(v_j, w_j) = \kappa_j > 0$, all cross-pair symplectic products zero, and all g -lengths one. Divide both vectors of pair j by $\sqrt{\kappa_j}$. Their symplectic product becomes one and their squared g -lengths become $\lambda_j = \kappa_j^{-1}$. Order the pairs with the sign convention of (F2). In these coordinates the dual form has reciprocal coefficients, so (F2) gives $h_g = \max_j \lambda_j$. This uses only finite-dimensional spectral decomposition. Unitary covariance transports (F31) and the form inequality. Enlarging the diagonal form to λe preserves the derivative bounds, since its unit directions are smaller. Thus it suffices to prove the assertion under (F25).

Induction. In spatial dimension zero the quantization is a nonnegative scalar. Assume the constant-metric theorem in dimension $n - 1$. A nonnegative symbol in dimension n that is independent of ξ_1 acts, at each fixed x_1 , as a Weyl operator in the other $n - 1$ variables. Its remaining derivatives obey the same constant-metric bounds, uniformly in the parameter x_1 . Apply the inductive estimate and integrate in x_1 . The identity follows first from the Weyl kernel on Schwartz functions, where Fourier inversion in ξ_1 produces the delta function in that coordinate; the quadratic-form estimate then follows by Fubini. The same conclusion holds for a symbol independent of any fixed real phase direction. Indeed an orthogonal symplectic map sends a unit vector in that direction to the ξ_1 direction: complete the pair v, Jv to orthonormal symplectic pairs in its invariant orthogonal complement. This map preserves e , and its unitary covariance preserves the lower bound.

Apply the adaptive construction (F26)–(F30). If $H_\nu = 1$, the localized symbol has uniformly bounded derivatives through the required order in $S(1, e)$. Section 7 of When a moving symbol scale controls an operator immediately bounds its entire operator norm. This treats the floor of the adaptive scale without subtracting a large constant from a nonnegative function.

If $H_\nu < 1$, the rescaled function (F27) satisfies the normalized splitting lemma. Write it as $v(z_\perp) + q(z)^2$ on the larger cutoff ball. Choose a nonnegative smooth cutoff in the transverse variables, equal to one on the projection of $\text{supp } \chi$, supported inside the domain of v . Multiplying v by it and extending by zero gives a global smooth nonnegative function $\tilde{v}$ independent of the same direction. Define

$$\begin{aligned} \chi_\nu(Y) &= \chi(\sqrt{H_\nu}(Y - X_\nu)), \\ b_\nu(Y) &= H_\nu^{-2} \tilde{v}(\sqrt{H_\nu}(Y - X_\nu)), \\ c_\nu(Y) &= H_\nu^{-1} (\chi q)(\sqrt{H_\nu}(Y - X_\nu)). \end{aligned} \quad (\text{F32})$$

The real function χq is extended by zero from a domain strictly larger than its support. We have the global identity

$$a_\nu = \chi_\nu^2 b_\nu + c_\nu^2. \quad (\text{F33})$$

The derivative bounds from the splitting lemma give uniform seminorms, through order $N_n - 2$, in

$$\chi_\nu \in S(1, H_\nu e), \quad b_\nu \in S(H_\nu^{-2}, H_\nu e), \quad c_\nu \in S(H_\nu^{-1}, H_\nu e). \quad (\text{F34})$$

Since b_ν is independent of one direction, induction yields $b_\nu^w \geq -C$ uniformly. Its symbol bounds may have a fixed constant instead of one; division by that constant before using induction and multiplication afterward give the same conclusion with another fixed C .

Expand the two products in

$$T_\nu = \chi_\nu^w b_\nu^w \chi_\nu^w + (c_\nu^w)^2. \quad (\text{F35})$$

The first-order terms cancel by (F23) and $\{c_\nu, c_\nu\} = 0$. Their remaining symbol weight is $H_\nu^2 H_\nu^{-2} = 1$. Constant-metric continuity therefore bounds $\|a_\nu^w - T_\nu\|$ independently of ν, λ . The cutoff operator χ_ν^w is uniformly bounded. Since c_ν is real, its contribution is a square, so

$$\langle a_\nu^w u, u \rangle \geq -C \|\chi_\nu^w u\|^2 - C' \|u\|^2 \geq -C'' \|u\|^2. \quad (\text{F36})$$

These bounds also hold in the previously treated $H_\nu = 1$ branch. Apply (F24) for the adaptive metric to obtain the desired bound for a .

Finite derivative dependence and approximation. Every use of the product, remainder, partition and continuity estimates above asks for finitely many input seminorms. The structural constants of the adaptive metric depend only on the fourth-derivative normalization and dimension. Let J_n exceed all input orders needed to bound the order-two product errors, their required output seminorms, the cutoff operators and the squared-localization error in this proof. These are finite indices supplied by the continuity estimates, independent of a, λ . Choose

$$N_0 = 0, \quad N_n \geq \max\{4, J_n + 2, N_{n-1} + 2\}. \quad (\text{F37})$$

The two additional derivatives pay for the splitting lemma. Formula (F28) bounds all the rescaled input derivatives up to this chosen order. Thus neither the induction nor the summation asks for infinitely many uniformly bounded derivatives.

To justify the symbol calculus when only these finite global bounds are assumed, first multiply a by $\zeta(X/R)^2$, with a fixed nonnegative compactly supported cutoff equal to one near zero. Take $R \geq \lambda^{-1/2}$. The product rule and (F25) give the same finite bounds up to a fixed constant independent of R, λ , because a derivative on the cutoff contributes at most $\lambda^{1/2}$. A fixed normalization absorbs that constant. The compactly supported symbols have every seminorm finite, so all preceding calculus operations are legitimate. Their adaptive structural constants and all constants actually used are uniform. Let $R \rightarrow \infty$ at fixed λ, a . The cutoffs converge locally smoothly to one, with a common polynomial bound, and pairing against the Schwartz Wigner function passes to the limit. This proves the theorem as stated. $\square$

8. Gluing the constant estimates for a variable metric

Scalar Fefferman–Phong theorem. Let g be permissible and let $0 \leq a \in S(h^{-2}, g)$ be scalar. Then the second line of (F5) holds.

Proof. Construct a real squared partition for g . Choose nonnegative cutoffs ψ_ν in slightly larger metric balls, equal to one on $\text{supp } \phi_\nu$, with uniform $S(1, g)$ bounds. Put $a_\nu = \psi_\nu a$. The functions are nonnegative. On each support, slow variation compares g with $g_\nu = g_{X_\nu}$, and h with $h_\nu = h(X_\nu)$. The product rule gives

$$|a_\nu|_{k, g_\nu} \leq C_k h_\nu^{-2} \quad (\text{F38})$$

globally, since the function vanishes outside its support. The constant metric g_ν has Planck parameter $h_\nu \leq 1$. Divide by $\max_{k \leq N_n} C_k$, apply Section 7 with $\lambda = h_\nu$, and multiply back. The lower bound is uniform in ν . Section 5 applies because $\phi_\nu^2 a_\nu = \phi_\nu^2 a$, so (F24) proves the theorem. This application uses the completed constant-metric induction, never the theorem currently being proved. $\square$

9. The classical endpoint and a change of quantization

For $0 \leq \delta < \rho \leq 1$, take

$$g_{x,\xi} = \langle \xi \rangle^{2\delta} |dx|^2 + \langle \xi \rangle^{-2\rho} |d\xi|^2, \quad h = \langle \xi \rangle^{\delta-\rho}. \quad (\text{F39})$$

The metric is permissible by Section 8 of Two measuring scales, one Weyl product, and $h^{-2} = \langle \xi \rangle^{2(\rho-\delta)}$. Therefore

$$0 \leq a \in S_{\rho,\delta}^{2(\rho-\delta)} \implies a^w \geq -C. \quad (\text{F40})$$

The same conclusion, with another constant, holds for the symmetric part $a(x, D) + a(x, D)^*$. Indeed the Weyl symbol of the left quantization has expansion

$$b = a + \frac{i}{2} \sum_j \partial_{x_j} \partial_{\xi_j} a + r, \quad r \in S_{\rho,\delta}^0, \quad (\text{F41})$$

with the sign fixed by (F1); for instance the left symbol $x\xi$ has Weyl symbol $x\xi + i/2$. The first correction is purely imaginary since a is real. Thus the symmetric part has Weyl symbol $b + \bar{b} = 2a + 2\operatorname{Re} r$, and the last term is bounded on L^2 . Apply (F40) to $2a$. The stated corollary retains the strict inequality $\delta < \rho$; no type $(1, 1)$ endpoint is being added by this argument.

Editorial endpoint extension. At $\rho = \delta = r$, $0 \leq r < 1$, the original metric and Planck parameter in (F39) are

$$g_{x,\xi} = \langle \xi \rangle^{2r} |dx|^2 + \langle \xi \rangle^{-2r} |d\xi|^2, \quad h(X) = 1, \quad h^{-2} = 1. \quad (\text{F51})$$

This metric equals its symplectic dual. It is slowly varying: a displacement of sufficiently small squared g_X -length has $|\eta - \xi| \leq \frac{1}{2} \langle \xi \rangle^r \leq \frac{1}{2} \langle \xi \rangle$, so $\langle \eta \rangle$ lies between $\langle \xi \rangle/2$ and $3\langle \xi \rangle/2$. For temperateness set $t = g_Y(X - Y)$. Then $|\xi - \eta| \leq t^{1/2} \langle \eta \rangle^r$, so $\langle \xi \rangle / \langle \eta \rangle \leq 1 + t^{1/2}$. If $\langle \xi \rangle \geq \langle \eta \rangle/2$, the inverse ratio is at most two. Otherwise, when $\langle \eta \rangle \geq 2$, the Lipschitz estimate gives $|\xi - \eta| \geq \langle \eta \rangle/2$, and therefore $t \geq \langle \eta \rangle^{2(1-r)}/4$. When $\langle \eta \rangle < 2$, the inverse ratio is already at most two. These inequalities bound both coefficient ratios of g_X/g_Y by $C_r(1+t)^{r/(1-r)}$; at $r = 0$ the metric is constant and the exponent is zero. This proves the original metric's temperateness, with the distance based at Y .

The directional seminorms of $S(1, g)$ are equivalent, at each fixed finite order, to the coordinate seminorms of $S_{r,r}^0$. Indeed the coordinate direction lengths are $\langle \xi \rangle^r$ in position and $\langle \xi \rangle^{-r}$ in frequency; expansion of each multilinear derivative in the corresponding orthonormal coordinate directions gives the finite dimensional constants. Thus (B26) supplies the bounded Weyl operator and its lower bound on this same metric. The conclusion for $a \in S_{r,r}^0$ is a bounded-operator lower bound. For $0 < r < 1$, the complete band-summation theorem (P14) of Positivity through a moving family of scalar probes, with its original parameters $\rho = \delta = r$ and symbol and Sobolev orders zero, also gives the left-operator bounds

$$\|a(x, D)\| \leq Cp_{0,L}(a), \quad \operatorname{Re} \langle a(x, D)u, u \rangle \geq -Cp_{0,L}(a)\|u\|^2. \quad (\text{F52})$$

For $r = 0$, the packet proof (P17)–(P18) gives the same estimate directly, since every coordinate derivative through its chosen finite order is bounded; its kernel marginal integrals converge. Equivalently, (B26) applies on the original constant metric $g = e$, with the quantization conversion estimates already proved. Both proofs retain the original symbol and its finite seminorms.

The adjoint has the same operator norm, so the symmetric part obeys $\langle (a(x, D) + a(x, D)^*)u, u \rangle \geq -2Cp_{0,L}(a)\|u\|^2$. These bounds do not require $a \geq 0$. Thus the equality endpoint extends the lower-bound conclusion as a zero-order boundedness statement, with no decreasing remainder weight and no positivity improvement. The type $(1, 1)$ corner remains excluded: the proof still requires $r < 1$ in its band remainder summation. This does not alter the nonnegative inverse-square result or its sharp exponent.

A square-root proof retaining the original metric

$$a \geq 0, \quad a \in S(h^{-1}, g), \quad m = a + 1$$

$$G = g/(hm), \quad \mu = (hm)^{-1} \geq (1 + p_0(a; h^{-1}, g))^{-1} > 0$$

$$h_G = 1/m, \quad b = \sqrt{m}, \quad b \# b = m + r$$

$$r \in S(mh_G^2, G) = S(m^{-1}, G) \subset S(1, G)$$

$$\langle a^w u, u \rangle = \|b^w u\|^2 - \|u\|^2 - \langle r^w u, u \rangle$$

The bounded remainder gives the lower bound; the original g is retained.

F7, F11a and F13 prove the displayed identities. A56–A58 prove temperateness for every fixed positive lower bound.
This diagram records a complete proof, not a changed symbol.

Figure 1: The adapted square on the original metric

The diagram keeps the original g, h, a and every factor in (F7), (F11a), (F13). Equations (A56)–(A58) prove the conformal comparison used here. This calculation uses the original conformal factor without a metric replacement.

10. Why the inverse square is the last uniform power

In one spatial dimension let $b(x, \xi) = x\xi$ and $p = b^2$. The exact polynomial Weyl product is

$$b \# b = b^2 + \frac{1}{4}, \quad \langle p^w u, u \rangle = \|b^w u\|^2 - \frac{1}{4} \|u\|^2. \quad (\text{F42})$$

For a direct check, $b^w = -i(x\partial_x + 1/2)$, whereas $(x^2\xi^2)^w = -x^2\partial_x^2 - 2x\partial_x - 1/2$. Squaring the first expression gives the second plus $1/4$.

Choose a nonzero real $\eta \in C_c^\infty((0, 1))$ and, for $L > 1$, set

$$u_L(x) = x^{-1/2}\eta((\log x)/L) \quad (x > 0), \quad u_L(x) = 0 \quad (x \leq 0). \quad (\text{F43})$$

Each function is smooth with compact support away from zero. Substituting $t = \log x$ gives

$$\|u_L\|^2 = L\|\eta\|^2, \quad \|b^w u_L\|^2 = L^{-1}\|\eta'\|^2. \quad (\text{F44})$$

For one fixed sufficiently large L , (F42) is strictly negative. This is an independently chosen logarithmic test family; it uses no spectral assertion about the dilation generator.

Let $\theta \geq 0$ be smooth, compactly supported on phase space, and equal to one near zero. Put $p_0(X) = \theta(X)p(X) \geq 0$. Then

$$\lambda^{-2}p_0(\sqrt{\lambda}X) = \theta(\sqrt{\lambda}X)p(X) \longrightarrow p(X) \quad \text{in } \mathcal{S}'(\mathbb{R}^2), \quad (\text{F45})$$

as $\lambda \downarrow 0$: the convergence is pointwise and bounded by a fixed quartic polynomial, so dominated convergence applies against each Schwartz function. Pairing with the Wigner function of the fixed u_L yields

$$\lambda^{-2}\langle p_0(\sqrt{\lambda}\cdot)^w u_L, u_L \rangle \longrightarrow \langle p^w u_L, u_L \rangle < 0. \quad (\text{F46})$$

Fix $s > 2$, and define $a_\lambda(X) = \lambda^{-s} p_0(\sqrt{\lambda}X)$, with constant metric $g_\lambda = \lambda e$. Then $h_{g_\lambda} = \lambda$, and

$$p_k(a_\lambda; \lambda^{-s}, g_\lambda) = \sup_X |p_0^{(k)}(X)|. \quad (\text{F47})$$

Thus every relevant symbol seminorm is uniform, but the quadratic form in (F46), multiplied by λ^{2-s} , tends to minus infinity. A fixed positive rescaling of p_0 can make any prescribed finite list of these seminorms at most one; it does not change the divergence. No uniform theorem of the constant-metric form (F31) can replace λ^{-2} by λ^{-s} .

One can also put this obstruction into a single variable metric and a single symbol. This avoids interpreting a family of metrics as though it were one fixed operator. Still in one phase plane, take

$$g_X = \langle X \rangle^{-1} e, \quad h(X) = \langle X \rangle^{-1}, \quad R_j = 2^{2j+j_0}, \quad X_j = (R_j, 0), \quad A(X) = \sum_{j \geq 1} R_j^s p_0((X - X_j)/\sqrt{R_j}). \quad (\text{F48})$$

Choose the fixed integer j_0 sufficiently large. The summands then have disjoint supports, of radius at most a fixed multiple of $\sqrt{R_j}$, so the sum is locally finite and nonnegative. On each support $\langle X \rangle$ is comparable to R_j ; differentiating the summand proves $A \in S(h^{-s}, g)$. Slow variation of this metric follows from $|\langle X \rangle - \langle Y \rangle| \leq |X - Y|$, and (F29), with $H = \langle X \rangle^{-1}$, proves symplectic temperateness. Its uncertainty inequality holds.

Let v_j be the unitary phase translation of the one fixed test function u_L to X_j . The contribution of the j -th summand to its quadratic form is at most $-cR_j^{s-2}$ for large j , by (F46). Every other contribution is negligible in absolute value. Here is a quantitative justification. The Wigner function of u_L is Schwartz, and the kernel formula (F1) pairs it with the symbol, up to the fixed factor $(2\pi)^{-1}$. For any integer M , the absolute contribution of the k -th summand with $k \neq j$ is at most

$$C_M R_k^{s+1} (1 + |X_k - X_j|)^{-M}. \quad (\text{F49})$$

The power R_k in the volume factor is the area of its phase-space support. The chosen growth of the centers makes that support's radius negligible relative to the distance between different centers. For $k < j$, the sum of (F49) is at most $C_M R_j^{-M} R_{j-1}^{s+1}$; for $k > j$, it is at most $C_M \sum_{k > j} R_k^{s+1-M}$. Taking $M > s + 3$ makes both bounds tend to zero. The fixed L^2 norm of v_j therefore accompanies a quadratic form tending to minus infinity. Thus the general theorem with h^{-s} , $s > 2$, fails even for one permissible metric and one nonnegative scalar symbol.

The example concerns scalar Weyl positivity and the uniform exponent. It neither gives an operator-valued Fefferman–Phong theorem nor supplies the separate matrix counterexamples needed to classify that setting.

11. Examples separating the assumptions

Subtracting a constant from a positive square. The operator $(xD + Dx)^2/4$ is nonnegative on Schwartz functions, and its Weyl symbol is $x^2\xi^2 + 1/4$. Subtracting $1/4$ leaves the nonnegative symbol $x^2\xi^2$, but its operator has negative test forms by (F42)–(F44). Positivity of a function and positivity of its quantization are two different statements even for a polynomial.

A singular zero set without a positive Hessian. The function $f(s, y) = s^4 + y^6$ has both value and Hessian zero at the origin. The adaptive scale there is at its floor. The proof does not invoke an implicit critical graph with a nonzero Hessian at that point. It uses an ordinary bounded-symbol estimate on that scale, while nearby points with a larger value or Hessian can use the splitting argument. This is why the theorem includes arbitrarily degenerate zeros.

A classical symbol at a nonclassical derivative scale. Let $\rho = 3/4$, $\delta = 1/4$, and let $F \geq 0$ be smooth on $\mathbb{R}$ with every derivative bounded. In one dimension,

$$a(x, \xi) = \langle \xi \rangle F(x \langle \xi \rangle^{1/4}) \chi(x) \quad (\text{F50})$$

belongs to $S_{3/4, 1/4}^1$ if $\chi \geq 0$ is smooth with compact support. To check the frequency derivatives, every derivative of the composed argument produces a factor bounded on $\text{supp } \chi$ by $C \langle \xi \rangle^{-3/4}$; subsequent derivatives improve this bound. An x derivative costs at most $\langle \xi \rangle^{1/4}$. The product rule gives the asserted class. Since $1 = 2(\rho - \delta)$, (F40) applies. This example illustrates the derivative accounting; the theorem is not restricted to such separated symbols or to compact support in x .

12. Problems and complete solutions

Problem 1. For a constant metric $g = \alpha dx^2 + \beta d\xi^2$ on one phase plane, compute its dual and Planck parameter. Find a symplectic dilation making its two coefficients equal.

Solution. Formula (F2) gives $g^\sigma = \beta^{-1} dx^2 + \alpha^{-1} d\xi^2$, so $h^2 = \alpha\beta$. Under $(x, \xi) = (cy, c^{-1}\eta)$, the coefficients become αc^2 and βc^{-2} . Taking $c = (\beta/\alpha)^{1/4}$ makes both equal to $\sqrt{\alpha\beta} = h$. The map preserves $dx \wedge d\xi$; its unitary action on functions is the corresponding normalized dilation. Thus arbitrary eccentricity does not enter the constant-metric lower bound.

Problem 2. Why does a single square-root argument applied naively to $a + 1$ at size h^{-2} fail to prove the second line of (F5)?

Solution. For a general nonnegative symbol of that size, the first-derivative positivity estimate is $|a'| \lesssim \sqrt{a} h^{-1} g^{1/2}$. The natural conformal metric that makes $a + 1$ its own weight is then $G = (a + 1)^{-1} h^{-2} g$. Its Planck parameter is $h_G = (a + 1)^{-1} h^{-1}$, which can exceed one near a zero of a . The Weyl square remainder also has weight $(a + 1)h_G^2 = (a + 1)^{-1} h^{-2}$, which need not be bounded. The scalar splitting and dimensional reduction resolve precisely this failure; a formal square root alone does not.

Problem 3. Verify the cancellation in (F23) and state the remaining weight when a has weight h^{-2} .

Solution. The Leibniz rule gives $\{\phi a, \phi\} = \phi\{a, \phi\} + a\{\phi, \phi\} = -\phi\{\phi, a\}$. Scalar multiplication commutes, so it cancels $\{\phi, a\}\phi$. The first surviving terms contain two symplectic contractions. Their weight is $h^{-2}h^2 = 1$, both for the direct second correction and for the composition of two first corrections. That is why the localization error is bounded in L^2 .

Problem 4. Suppose $a^{(k)}$ satisfies (F25) and H is defined by (F26). Explain both inequalities $\lambda \leq H \leq 1$, and compute the rescaled k -th derivative in (F27).

Solution. The maximum defining H^{-1} is at least one. Its other two entries are at most λ^{-1} , by the order-zero and order-two bounds in (F25); since $\lambda \leq 1$, the maximum is at most λ^{-1} . Thus $\lambda \leq H \leq 1$. The chain rule gives $f_X^{(k)} = H^{2-k/2} a^{(k)}(X + H^{-1/2}z)$. For $k \geq 4$, its norm is at most $(\lambda/H)^{(k-4)/2} \leq 1$. The lower derivative bounds instead use nonnegativity and the jet lemma; (F25) alone would give negative powers of λ/H there.

Problem 5. For the logarithmic family (F43), prove the negative-form condition and explain why one fixes L before letting λ tend to zero.

Solution. Equations (F42) and (F44) give $\langle p^w u_L, u_L \rangle = L^{-1} \|\eta'\|^2 - (L/4) \|\eta\|^2$, negative when $L^2 > 4 \|\eta'\|^2 / \|\eta\|^2$. Fix such an L . Its Wigner function is now one fixed Schwartz test function, so the distributional convergence in (F45) gives (F46) directly. Letting the test function change with λ would require a separate uniform convergence estimate; none is needed for this counterexample.

Problem 6. A localized splitting has a profile v independent of one phase direction but supported only on a transverse ball. Describe a global extension preserving the direction independence and nonnegativity, and explain why cutting off in all phase directions at that step is unsuitable.

Solution. Project the compact support of χ onto the orthogonal complement of the constant direction. This projection lies strictly inside the transverse domain on which v is defined. Choose a nonnegative transverse cutoff equal to one near this projection and supported inside that domain. Multiply v by it and extend by zero in the transverse variables; leave the constant direction unrestricted. The extension is smooth, nonnegative and independent of that direction, and satisfies $\chi^2 \tilde{v} = \chi^2 v$. The later factor χ supplies the remaining localization in (F33). A cutoff varying in the missing direction would destroy the parameter decomposition used in dimension induction.

13. Exact source comparisons and visible corrections

The following is a separate editorial comparison, keeping the preceding scalar theorem and proof intact. It compares Wen Deng, *Structure constants of the Weyl calculus*, arXiv:1109.4793v1, with the original course formulas (F1)–(F5) and (F42). The source’s own wording and valid theorem remain identifiable. The two source-text corrections below are not errors in the course’s independent scalar proof. The original author files have not been edited. There is no novelty claim.

The original source uses a measurable positive quadratic metric, the symplectic form $\sigma = \sum_j d\xi_j \wedge dx_j$, a continuous confined-symbol partition and imported Wiener and biconfinement theorems. The comparisons below apply on that common measurable-metric scope. They do not use the source to certify a larger nonmeasurable scope or to supply the missing proofs in its cited books.

13.1. Every Fourier and metric factor in the comparison

Keep the original course symbol $a(x, \xi)$, its original metric g , original symplectic form and Lebesgue measure. Put $c = 2\pi$ and introduce only the comparison map

$$L(x, \eta) = (x, c\eta), \quad \det L = c^n, \quad a_D = a \circ L. \quad (\text{SD1})$$

For a Schwartz symbol and test function, substitution $\xi = c\eta$ in the absolutely convergent source quantization gives

$$\begin{aligned} (a_D)_D^w u(x) &= \iint e^{ic(x-y)\cdot\eta} a((x+y)/2, c\eta) u(y) dy d\eta \\ &= c^{-n} \iint e^{i(x-y)\cdot\xi} a((x+y)/2, \xi) u(y) dy d\xi = a^w u(x). \end{aligned} \quad (\text{SD2})$$

For a tempered symbol the same identity is an equality of distributional kernels. Here is the exact extension, rather than an ordinary integral assertion: the partial Fourier transform is a continuous bijection of the Schwartz space and its distribution dual, and the coordinate change $(x, y) \mapsto ((x+y)/2, x-y)$ is invertible. They define both kernel maps continuously on tempered symbols. The distribution pullback under L is $\langle a_D, \varphi \rangle = c^{-n} \langle a, \varphi \circ L^{-1} \rangle$. Applying these continuous maps to this identity gives precisely (SD2), including its c^{-n} . Equivalently one may test against the Schwartz Wigner function of two Schwartz functions; the same substitution has the same determinant. The complete Fourier, linear-substitution and distribution maps are already proved in the course prerequisites cited before (F2).

For all vectors T, S , $\sigma(LT, LS) = c\sigma(T, S)$. Write $\tilde{g} = L^*g$, meaning $\tilde{g}_Y(T) = g_{LY}(LT)$. Its original

symplectic dual is

$$\begin{aligned}\tilde{g}_Y^\sigma(T) &= \sup_{S \neq 0} \frac{\sigma(T, S)^2}{g_{LY}(LS)} = c^{-2} g_{LY}^\sigma(LT), \\ \hat{g} &= c^{-1} L^* g, \quad \hat{g}_Y^\sigma = c^{-1} L^*(g^\sigma)_Y.\end{aligned}\tag{SD3}$$

The second line uses the exact rule $(bq)^\sigma = b^{-1}q^\sigma$ for $b > 0$, proved directly from the quotient defining the dual. Thus $\hat{g} \leq \hat{g}^\sigma$ is equivalent to the original $g \leq g^\sigma$. The map has not replaced the original metric in the course calculation.

The original course quantity is $h(X)^2 = \sup_{T \neq 0} g_X(T)/g_X^\sigma(T)$. The source quantity is the reciprocal ratio, not the same quantity:

$$h_{\hat{g}}(Y) = h(LY), \quad \lambda_{\hat{g}}(Y) = \inf_{T \neq 0} \left(\frac{\hat{g}_Y^\sigma(T)}{\hat{g}_Y(T)} \right)^{1/2} = h(LY)^{-1}.\tag{SD4}$$

The positive ratios attain finite positive extrema on a Euclidean unit sphere; reciprocating therefore changes their supremum into the reciprocal infimum. This also proves the identity at every base point without an exceptional zero or infinite value.

For the exact order- k directional seminorm, with the original weight h^{-2} , the full comparison is

$$p_k(a_D; (h \circ L)^{-2}, \hat{g}) = c^{k/2} p_k(a; h^{-2}, g).\tag{SD5}$$

A $\hat{g}$ -unit vector maps to a vector of original g -length $\sqrt{c}$; each of the k arguments contributes that factor. Conversely every such original vector is obtained by the inverse map. The weight has unchanged value at the corresponding point. For the source maximum through order l , the exact expression is $\max_{0 \leq k \leq l} c^{k/2} p_k(a; h^{-2}, g)$, bounded by $c^{l/2} \max_{0 \leq k \leq l} p_k(a; h^{-2}, g)$. Do not substitute that upper bound for the exact maximum.

Slow variation retains its full radius: if g has comparison constant C_0 for original squared distance at most C_0^{-1} , then the same comparison applies when $\hat{g}_Y(Y - Z) \leq (cC_0)^{-1}$. In the original distance $g_{LY}^\sigma(L(Y - Z)) = c\hat{g}_Y^\sigma(Y - Z)$. A one-base temperateness power N consequently retains the factor c^N as an upper bound when expressed with $1 + \hat{g}_Y^\sigma(Y - Z)$. Section2 below gives the exact comparison with the source's harmonic-mean distance. Together (SD2)–(SD5) identify the lower-bound conclusions on their common scope with finite structural and symbol constants retained.

The polynomial correction is equally explicit. With the original $D = -i\partial_x$, let $K = (xD + Dx)/2$. Original (F42) gives $(x^2\xi^2)^w = K^2 - 1/4$. The source operator for the unscaled coordinate symbol $x\eta$ is K/c , and

$$(x^2\eta^2)_D^w = c^{-2}(K^2 - 1/4) = (K/c)^2 - \frac{1}{4c^2} = (K/c)^2 - \frac{1}{16\pi^2}.\tag{SD6}$$

Here $(x^2\xi^2) \circ L = c^2 x^2 \eta^2$ and linearity supply the factor c^{-2} ; no square constant has been deleted or identified with a differently scaled symbol.

13.2. The exact two-way temperateness map

Let $Q_X = g_X^\sigma$. Suppose the original (F3) holds in its actual orientation:

$$g_Y \leq C g_X (1 + Q_Y(X - Y))^N, \quad C \geq 1, \quad N \geq 0.\tag{SD7}$$

For $d = X - Y$ minimize $Q_X(u) + Q_Y(v)$ under $u + v = d$. In the original symmetric positive matrices the unique critical point and minimum are

$$\begin{aligned}u &= (Q_X + Q_Y)^{-1} Q_Y d, \quad v = d - u, \\ A + B &:= Q_X(u) + Q_Y(v) = d^T (Q_X^{-1} + Q_Y^{-1})^{-1} d = \frac{1}{2} (Q_X \wedge Q_Y)(d), \\ Q_X \wedge Q_Y &= 2(Q_X^{-1} + Q_Y^{-1})^{-1}.\end{aligned}\tag{SD8}$$

To prove the minimum, expand $u^T Q_X u + (d-u)^T Q_Y (d-u)$, complete the square with matrix $Q_X + Q_Y$, and retain the remaining matrix $Q_Y - Q_Y(Q_X + Q_Y)^{-1}Q_Y$. It equals $Q_Y(Q_X + Q_Y)^{-1}Q_X$. Its inverse is $Q_X^{-1}(Q_X + Q_Y)Q_Y^{-1} = Q_Y^{-1} + Q_X^{-1}$; this proves (SD8) without commuting the original matrices.

Put $Z = X - u = Y + v$. The orientation of (SD7) gives $g_X \leq Cg_Z(1+A)^N$. Duality gives $Q_Z \leq CQ_X(1+A)^N$. A second application gives

$$g_Z \leq Cg_X(1 + Q_Z(u))^N \leq Cg_X[1 + CA(1+A)^N]^N.$$

Also $g_Y \leq Cg_Z(1+B)^N$. Since $1 + CA(1+A)^N \leq (1+C)(1+A)^{N+1}$, multiplication gives the complete source-distance bound

$$\begin{aligned} g_Y &\leq C^2(1+C)^N g_X(1+A+B)^{N^2+2N} \\ &\leq C^2(1+C)^N g_X[1 + (Q_X \wedge Q_Y)(X-Y)]^{N^2+2N}. \end{aligned} \quad (\text{SD9})$$

The source orientation with g_X on the left follows by interchanging X, Y ; the mean is symmetric. Conversely choose $u = d, v = 0$ in the minimum: $(Q_X \wedge Q_Y)(d) \leq 2Q_X(d)$, and choose $u = 0, v = d$ to obtain the separate bound by $2Q_Y(d)$. Therefore a harmonic-mean temperateness bound with constant C_D and power N_D implies (SD7) with $C_D 2^{N_D}$ and N_D . This proves both maps, including the case $N = 0$.

13.3. The corrected quadratic-cutoff derivatives and their full receiver

In source Section2, original TeX line223 misses a factor2. Fix the original base Y , positive quadratic form g_Y , its associated inner product, original radius $r > 0$, and the actual smooth nonincreasing cutoff χ_0 , with $\chi_0 = 1$ for arguments at most $1/2$, $\chi_0 = 0$ for arguments at least1, and $0 \leq \chi_0 \leq 1$. Write $z = X - Y$, $q_Y(X) = r^{-2}g_Y(z)$, and $\omega_Y(X) = \chi_0(q_Y(X))$. The exact first derivative is

$$D\omega_Y(X)[T] = 2r^{-2}\chi'_0(r^{-2}g_Y(X-Y))\langle X-Y, T \rangle_Y. \quad (\text{SD10})$$

Indeed $g_Y(z + tT) = g_Y(z) + 2t\langle z, T \rangle_Y + t^2g_Y(T)$. This is fixed by the original quadratic form; changing the meaning of its associated inner product would change the original data.

For every $k \geq 1$, the entire repeated-direction formula is

$$D^k\omega_Y(X)[T^k] = \sum_{p=\lceil k/2 \rceil}^k \frac{k! 2^{2p-k}}{(2p-k)!(k-p)!} \chi_0^{(p)}(r^{-2}g_Y(z)) r^{-2p} \langle z, T \rangle_Y^{2p-k} g_Y(T)^{k-p}. \quad (\text{SD11})$$

Taylor-expand the original χ_0 through order k in the increment $r^{-2}(2t\langle z, T \rangle_Y + t^2g_Y(T))$. The order- p power contributes to t^k only by choosing $2p-k$ linear factors and $k-p$ quadratic factors. Multiplying its coefficient by $k!$, including the Taylor $1/p!$, gives exactly (SD11). The Taylor remainder is $o(t^k)$, so this coefficient calculation proves the derivative even where some factors vanish. At order zero the expression is just $\chi_0(q_Y)$.

Every mixed direction is retained as well. If $\Pi_k^{1,2}$ denotes the set of partitions of $\{1, \dots, k\}$ whose blocks have size1 or2, then

$$\begin{aligned} D^k\omega_Y(X)[T_1, \dots, T_k] &= \sum_{\pi \in \Pi_k^{1,2}} \chi_0^{(|\pi|)}(q_Y(X)) \prod_{B \in \pi} D^{|B|}q_Y(X)[T_i : i \in B], \\ Dq_Y[T_i] &= 2r^{-2}\langle z, T_i \rangle_Y, \quad D^2q_Y[T_i, T_j] = 2r^{-2}\langle T_i, T_j \rangle_Y. \end{aligned} \quad (\text{SD12})$$

All derivatives of q_Y of order at least3 vanish. Repeated differentiation of the product adds the new index either as a singleton differentiating χ_0 , or to a singleton block differentiating Dq_Y . These choices

generate each new partition exactly once; adjoining it to a double block gives zero. This proves (SD12) by induction and proves the full mixed formula without omitting multiplicities.

For $k \geq 1$ define the actual finite coefficient sum

$$C_k(\chi_0) = \sum_{p=\lceil k/2 \rceil}^k \frac{k! 2^{2p-k}}{(2p-k)!(k-p)!} \|\chi_0^{(p)}\|_\infty. \quad (\text{SD13})$$

On the support of a nonzero term, $g_Y(z) \leq r^2$. Cauchy–Schwarz gives $|\langle z, T \rangle_Y| \leq r g_Y(T)^{1/2}$. Thus every original power in (SD11) gives

$$|D^k \omega_Y(X)[T^k]| \leq C_k(\chi_0) r^{-k} g_Y(T)^{k/2} \leq C_k(\chi_0) r^{-k} C_0^{k/2} g_X(T)^{k/2}, \quad r^2 \leq C_0^{-1}. \quad (\text{SD14})$$

The last comparison applies slow variation at base Y , on the very support just identified. In (SD12) each singleton contributes at most $2r^{-1} g_Y(T_i)^{1/2}$ and each double block at most $2r^{-2} g_Y(T_i)^{1/2} g_Y(T_j)^{1/2}$. The exact mixed bound is the sum over π , with coefficient $2^{|\pi|} \|\chi_0^{(|\pi|)}\|_\infty r^{-k} C_0^{k/2}$. Therefore the source's estimates with unspecified finite constants keep their stated order and support after the equality is corrected; the omitted2 is restored in the constants explicitly.

The complete continuous-partition receiver keeps its original measure:

$$\omega(X, r) = \int_{\mathbb{R}^{2n}} \omega_Y(X) |g_Y|^{1/2} dY, \quad \varphi_Y(X) = \omega_Y(X) / \omega(X, r). \quad (\text{SD15})$$

Its lower bound is $\omega(X, r) \geq C_0^{-2n} r^{2n} \int \chi_0(|Z|^2) dZ > 0$. To prove it, where the comparison integrand is nonzero, $g_X(X - Y) \leq r^2 / C_0 \leq C_0^{-1}$, so $g_Y(X - Y) \leq C_0 g_X(X - Y)$, $|g_Y|^{1/2} \geq C_0^{-n} |g_X|^{1/2}$, and monotonicity of χ_0 applies. Substitution $Z = C_0^{1/2} r^{-1} g_X^{1/2}(Y - X)$ has the full Jacobian $r^{2n} C_0^{-n} |g_X|^{-1/2}$, proving that bound. The integral is positive since $\chi_0 = 1$ on the ball $|Z|^2 \leq 1/2$.

On the original support, $g_X(X - Y) \leq C_0 r^2$ and $|g_Y|^{1/2} \leq C_0^n |g_X|^{1/2}$. Hence

$$\begin{aligned} \omega(X, r) &\leq C_0^{2n} r^{2n} |B_{2n}|, \\ |D^k \omega(X, r)[T^k]| &\leq C_k(\chi_0) C_0^{2n+k/2} r^{2n-k} |B_{2n}| g_X(T)^{k/2}. \end{aligned} \quad (\text{SD16})$$

Here $|B_{2n}|$ is the actual Euclidean unit-ball volume; it has not been absorbed into another object. Differentiation under the integral is justified even for the source's measurable metric. Fix X_0 , take $g_{X_0}(X - X_0) \leq \varepsilon^2 \leq C_0^{-1}$, and consider a nonzero cutoff derivative. Two slow comparisons give g_Y comparable with g_{X_0} by C_0^2 , and $g_{X_0}(Y - X_0) \leq 2C_0^2 r^2 + 2\varepsilon^2$. Thus the integration support lies in one finite ellipsoid, the determinant density is at most $C_0^{2n} |g_{X_0}|^{1/2}$, and (SD12) gives a fixed integrable bound for every prescribed derivative on that neighborhood. For fixed Y the integrand is smooth in X ; dominated convergence and the integral form of the difference quotient prove each derivative and its continuity. No differentiability of $Y \mapsto g_Y$ was used.

The positive lower bound permits the full reciprocal chain formula

$$D^k(\omega^{-1})[T_1, \dots, T_k] = \sum_{\pi \in \Pi_k} (-1)^{|\pi|} |\pi|! \omega^{-1-|\pi|} \prod_{B \in \pi} D^{|B|} \omega[T_i : i \in B], \quad (\text{SD17})$$

where Π_k contains all set partitions. Differentiating t^{-1} gives $(-1)^p p! t^{-p-1}$; the same partition induction proves this formula. All its bounds are finite by (SD16) and the lower bound. The full product rule for $\omega_Y \omega^{-1}$ therefore gives the original smooth partition bounds. Finally $\int \varphi_Y(X) |g_Y|^{1/2} dY = \omega(X, r) / \omega(X, r) = 1$, with its original support and measure unchanged. This supplies the entire receiver of the corrected derivative, rather than only a first-order observation.

13.4. The literal Wiener-domain defect and a proved ambient repair

Source lines 541–543 first restrict a to Schwartz functions and then assert inclusion of all bounded smooth symbols with bounded derivatives. The original constant $a = 1$ disproves that literal inclusion: it has every required bounded derivative, but $\sup_X |X_1 a(X)| = \infty$, so it is not Schwartz. It does not contradict the intended localized Fourier condition.

Keep the original dimension $d = 2n$, original compact smooth lattice cutoff χ_0 , original locally finite partition $\sum_{j \in \mathbb{Z}^d} \chi_0(X - j) = 1$, and original Fourier convention $\mathcal{F}_D f(\Xi) = \int e^{-2\pi i X \cdot \Xi} f(X) dX$. For a tempered distribution introduce the separate ambient class

$$\begin{aligned} \widetilde{\mathcal{A}} &= \{a \in \mathcal{S}'(\mathbb{R}^d) : \omega_a \in L^1(\mathbb{R}^d)\}, \\ \omega_a(\Xi) &= \sup_{j \in \mathbb{Z}^d} |\mathcal{F}_D(\chi_j a)(\Xi)|, \quad \chi_j(X) = \chi_0(X - j), \quad \|a\|_{\widetilde{\mathcal{A}}} = \int \omega_a(\Xi) d\Xi. \end{aligned} \quad (\text{SD18})$$

Each compactly supported distribution has a smooth Fourier transform of polynomial growth: differentiate its action on the smooth exponential multiplied by a fixed compact test cutoff equal to one on its support. The distribution's finite-order bound on that compact set proves the asserted polynomial estimate for every derivative. Thus the countable supremum in (SD18) is measurable and the definition is meaningful. This is explicitly an editorial ambient-domain repair, not a silent alteration of the source's declaration.

For $a = 1$, translation gives

$$\mathcal{F}_D(\chi_j)(\Xi) = e^{-2\pi i j \cdot \Xi} \mathcal{F}_D \chi_0(\Xi), \quad \omega_1 = |\mathcal{F}_D \chi_0|. \quad (\text{SD19})$$

To prove its integrability and the full finite-derivative inclusion, let $a \in C^{d+1}$ have all coordinate derivatives through that order bounded. Write

$$\begin{aligned} M_0 &= \|\chi_0\|_1 \|a\|_\infty, \\ M_{d+1} &= \max_{1 \leq i \leq d} \sum_{b=0}^{d+1} \binom{d+1}{b} \|\partial_i^b \chi_0\|_1 \|\partial_i^{d+1-b} a\|_\infty. \end{aligned} \quad (\text{SD20})$$

For $|\Xi| \leq 1$, the transform has absolute value at most M_0 , uniformly in j . For $|\Xi| > 1$, choose i with $|\Xi_i| \geq |\Xi|/\sqrt{d}$. The full product derivative and integration by parts give

$$(2\pi i \Xi_i)^{d+1} \mathcal{F}_D(\chi_j a) = \mathcal{F}_D \left(\sum_{b=0}^{d+1} \binom{d+1}{b} (\partial_i^b \chi_j)(\partial_i^{d+1-b} a) \right), \quad (\text{SD21})$$

and hence $\omega_a(\Xi) \leq (2\pi)^{-d-1} d^{(d+1)/2} M_{d+1} |\Xi|^{-d-1}$. Every derivative term and its multiplicity has been retained. The actual polar measure gives

$$\|a\|_{\widetilde{\mathcal{A}}} \leq |B_d| M_0 + (2\pi)^{-d-1} d^{(d+1)/2} |S^{d-1}| M_{d+1} \int_1^\infty r^{-2} dr = |B_d| M_0 + (2\pi)^{-d-1} d^{(d+1)/2} |S^{d-1}| M_{d+1}. \quad (\text{SD22})$$

In particular all $S_{0,0}^0$ symbols lie in the repaired ambient class, and taking $a = 1$ proves the finiteness in (SD19). The same proof holds for bounded weak derivatives through order $d+1$, using the already proved weak product rule and distributional integration by parts. It does not infer the imported fourth-derivative scalar lower-bound theorem from Fourier integrability alone.

For completeness the ambient repair has the actual functional properties needed to state its inclusion. Let $K = \text{supp } \chi_0$, let $J_K = \{m \in \mathbb{Z}^d : m \in K - K\}$, and put $N_K = |J_K| < \infty$. There are at most N_K

nonzero partition terms at any point: if one index j_0 meets that point, every other index differs from j_0 by an element of J_K . Fourier inversion in (SD18) makes each $\chi_j a$ a continuous bounded function, with norm at most $\|a\|_{\mathcal{A}}$. Summing the locally finite partition identifies a as a continuous bounded function and proves

$$\|a\|_\infty \leq N_K \|a\|_{\mathcal{A}}. \quad (\text{SD23})$$

If the latter quantity is zero every $\chi_j a$ vanishes, so the distribution a vanishes. Triangle inequality and homogeneity follow from the countable supremum and the integral; the original quantity is a norm.

For a Cauchy sequence a_l in this norm, (SD23) gives a uniform limit $a \in C_b$. Each $\chi_j a_l$ converges uniformly on its fixed compact support, so its Fourier transform converges pointwise to that of $\chi_j a$. At fixed l , the supremum of their pointwise limits is at most the lower limit of the suprema. Fatou's lemma gives $\|a_l - a\|_{\mathcal{A}} \leq \liminf_{m \rightarrow \infty} \|a_l - a_m\|_{\mathcal{A}}$. It follows that a belongs to the class and that $a_l \rightarrow a$ in its original norm. Thus this repaired ambient space is complete.

For a, b in it the product is their actual continuous-function product. In the localized identity $\chi_j ab = \sum_k (\chi_j a)(\chi_k b)$, only indices with $k - j \in J_K$ occur. Fourier transform of each product is convolution with no multiplier under $\mathcal{F}_D$. Each convolution is bounded by $\omega_a * \omega_b$; the complete finite sum therefore gives

$$\omega_{ab} \leq N_K (\omega_a * \omega_b), \quad \|ab\|_{\mathcal{A}} \leq N_K \|a\|_{\mathcal{A}} \|b\|_{\mathcal{A}}. \quad (\text{SD24})$$

Tonelli justifies the convolution integral and retains its full factor N_K . The original norm has not been rescaled to suppress that factor. These proofs establish bounded continuous representatives, completeness and continuous multiplication for the explicit ambient repair. The source's separate imported Wiener-based sharp fourth-derivative theorem and its imported biconfinement bounds remain distinct literature dependencies.

13.5. Precise receiving and historical scope

The accompanying diagram is an exact two-dimensional sample of Section3. Its original matrix is $G = \begin{pmatrix} 1/2 & 1/8 \\ 1/8 & 1/4 \end{pmatrix}$, radius $r = 1$, displacement $z = (1, 1/2)$, and direction $T = (4/5, -2/5)$. The positive first pivot and determinant $7/64$ prove positivity. Direct multiplication gives $g^\sigma = (64/7)g$, so this constant metric also satisfies uncertainty and slow variation with $C_0 = 1$. The original line has $g(z + tT) = 11/16 + (7/10)t + (7/25)t^2$ and derivative $7/10 + (14/25)t$. The displayed orange arrow is exactly $(9/20)T$, not a unit-vector replacement.

For this sample only, the diagram chooses $\chi_0(q) = 1 - \eta(2q - 1)$, where $h(s) = e^{-1/s}$ for $s > 0$, zero otherwise, and $\eta(s) = h(s)/(h(s) + h(1 - s))$. Every right derivative of h at zero is the limit of an exponential times a polynomial in s^{-1} , hence vanishes, since each power times $e^{-1/s}$ tends to zero. Thus h is smooth and flat there. The denominator is everywhere positive, and $\eta' = [h'(s)h(1 - s) + h(s)h'(1 - s)]/[h(s) + h(1 - s)]^2 \geq 0$. Consequently this particular cutoff is smooth, nonincreasing, equal to 1 for $q \leq 1/2$ and zero for $q \geq 1$. It is one admissible instance of the original cutoff hypotheses; (SD10)–(SD17) still hold for every original cutoff satisfying those hypotheses. The solid derivative curve uses the complete (SD10), while the dashed source expression is exactly half that value.

(SD1)–(SD9) prove the exact symbol, operator, metric, reciprocal-parameter and distance maps relating original (F1)–(F5) and (F42) to the cited source theorem. (SD10)–(SD17) restore the derivative equality and propagate it through every cutoff order and the full continuous-partition receiver. (SD18)–(SD24) disprove only the literal Schwartz-domain inclusion and supply the explicit repaired ambient space and the full finite-derivative inclusion, with all original Fourier and lattice factors.

No original course scalar theorem is replaced. Its original discrete cover, scalar splitting, dimensional induction, squared localization and logarithmic test remain the current independent proof. The exact version compared here is Deng1109.4793v1, submitted 22 September 2011; its complete TeX and included bibliography were read. Its imported book theorems have not been reconstructed from their original author TeX here. The two corrections are visible source notes, with their complete arguments above.

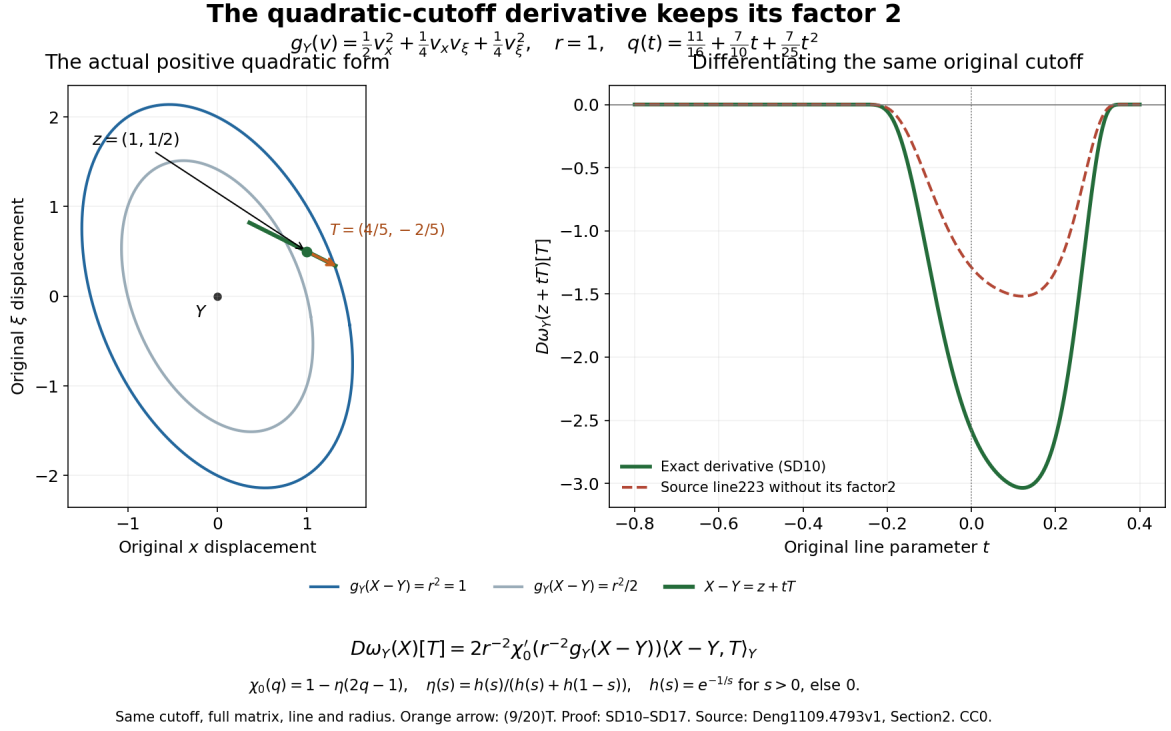


Figure 2: The original quadratic cutoff and its complete directional derivative

The left panel retains the complete matrix, both original ellipsoids, the displacement and direction. The right panel shows the exact derivative and the source expression with its missing factor2; the caption identifies the drawn arrow as (9/20)T. The exact sample and its smooth cutoff are proved in Section13.5.

References

Wen Deng's *Structure constants of the Weyl calculus*, arXiv:1109.4793v1, Theorem 4.1, states the general-metric scalar bound on its declared measurable-metric scope, with dependence on the metric's structural constants made explicit. Section13 proves the exact operator and metric comparison and records the two visible source corrections. Its proof uses a constant-metric result formulated through fourth derivatives in a Wiener algebra, followed by confined-symbol localization.

Nicolas Lerner's author-hosted *Metrics on the Phase Space*, chapter 2, Theorems 2.5.5 and 2.5.10 and Remarks 2.5.13–14, discusses the general-metric bound and finer fourth-derivative conditions. Its reciprocal Planck parameter and 2π -Fourier convention differ from (F1)–(F2). In particular constants in a polynomial Weyl-square identity must be converted before comparison.

Further questions

One direction for further work is an explicit, economical derivative count in (F37) for a specified class of metric structural constants. Another is to compare the sufficient pointwise condition $a \geq 0$ with averaged conditions for a lower bound. Formula (F42) and the logarithmic tests show that quantization can produce a bounded negative quadratic form from a nonnegative scalar symbol. Sufficient averaged conditions require a separate theorem; they do not follow by deleting nonnegativity from the splitting lemma. A third direction is to identify exactly where scalar multiplication and the scalar critical graph cease to work for matrix symbols. The general Hilbert-valued sharp lower bound belongs to its separate course unit, and an inverse-square improvement in that setting cannot be inferred from this scalar proof.